COMPACT ROTARY COLLAR / PHASE 0 LIMITS

Unchanged 5 V source path fails. No reduced assembly CAD or motion PASS is claimed.

Receptacle DC voltage at 1.5 A: module4.875 V; controller55 mOhm; termination20 mOhm

FPC: two8 mm traces, width-0.2 mm/copper-10%; hot copper80 C. Supplier qualification remains open.

Length / copper	20 C port	80 C port	Required receptacle minimum
0.8 m / 35 um	4.5916 V	4.5513 V	4.7500 V - FAIL
0.8 m / 70 um	4.6770 V	4.6569 V	4.7500 V - FAIL
1.0 m / 35 um	4.5488 V	4.4984 V	4.7500 V - FAIL
1.0 m / 70 um	4.6557 V	4.6305 V	4.7500 V - FAIL
1.1 m / 35 um	4.5275 V	4.4720 V	4.7500 V - FAIL
1.1 m / 70 um	4.6450 V	4.6173 V	4.7500 V - FAIL

Even optimistic1m/70um with zero terminations and25 C controller maximum gives4.72575 V.

181-angle spiral diagnostic: fixed radii25..30.5,6turns at neutral,0.16 allocated strip thickness.

Neutral length1046.165 mm; fixed-end winding changes length by +/-43.589 mm at +/-90 degrees.

Holding length instead forces outer radius32.912 mm at-90 degrees, outside the30.5 mm cavity.

This is a falsified simple model, not a proof that every clockspring is impossible. Tangent clamps remain unsolved.

One compensation direction: rotating local5.10 V +/-1% regulator BEFORE TPS25810.

Conditional minimum4.9365 V after switch/paths; about8.517 W input at90% assumed efficiency.

Requires an actual selected power stage, thermal/startup qualification and packing. No source architecture changed.

GCT six-contact source also lacks D+/D- required by USB-C R2.4 charger compatibility provision.

End-face nominal connector suggests near64 mm, but maximum dimensions + allowances screen64.51 mm.

Mains/PE remain stationary. No hardware tests, fixture/cable-fit claim, cycle rating or manufacturing release.